```
PS C:\Users\LENOVO> & C:/Users/LENOVO/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/LENOVO/OneDrive/Desktop/decode week 2/knn.py"

==============================
 IRIS DATASET INFORMATION
==============================

First 5 Rows:

   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  target
0                5.1               3.5                1.4               0.2       0
1                4.9               3.0                1.4               0.2       0
2                4.7               3.2                1.3               0.2       0
3                4.6               3.1                1.5               0.2       0
4                5.0               3.6                1.4               0.2       0

Dataset Shape: (150, 5)

Target Classes:
0 --> setosa
1 --> versicolor
2 --> virginica

==============================
 CHECKING NULL VALUES
==============================
sepal length (cm)    0
sepal width (cm)     0
petal length (cm)    0
petal width (cm)     0
target               0
dtype: int64

==============================
```

```
PS C:\Users\LENOVO> & C:/Users/LENOVO/AppData/Local/Python/pythoncore-3.14-64/python.exe "c:/Users/LENOVO/OneDrive/Desktop/decode week 2/knn.py"
==============================
 TRAIN TEST SPLIT
==============================
Training Data: (120, 4)
Testing Data : (30, 4)

Feature Scaling Completed Successfully!

==============================
 FINDING BEST K VALUE
==============================
K = 1 | Error Rate = 0.0333
K = 2 | Error Rate = 0.0667
K = 3 | Error Rate = 0.0667
K = 4 | Error Rate = 0.0667
K = 5 | Error Rate = 0.0667
K = 6 | Error Rate = 0.0667
K = 7 | Error Rate = 0.0333
K = 8 | Error Rate = 0.0667
K = 9 | Error Rate = 0.0333
K = 10 | Error Rate = 0.0333
K = 11 | Error Rate = 0.0333
K = 12 | Error Rate = 0.0333
K = 13 | Error Rate = 0.0333
K = 14 | Error Rate = 0.0333
K = 15 | Error Rate = 0.0333
K = 16 | Error Rate = 0.0333
K = 17 | Error Rate = 0.0333
K = 18 | Error Rate = 0.0333
K = 19 | Error Rate = 0.0333
K = 20 | Error Rate = 0.0333
```

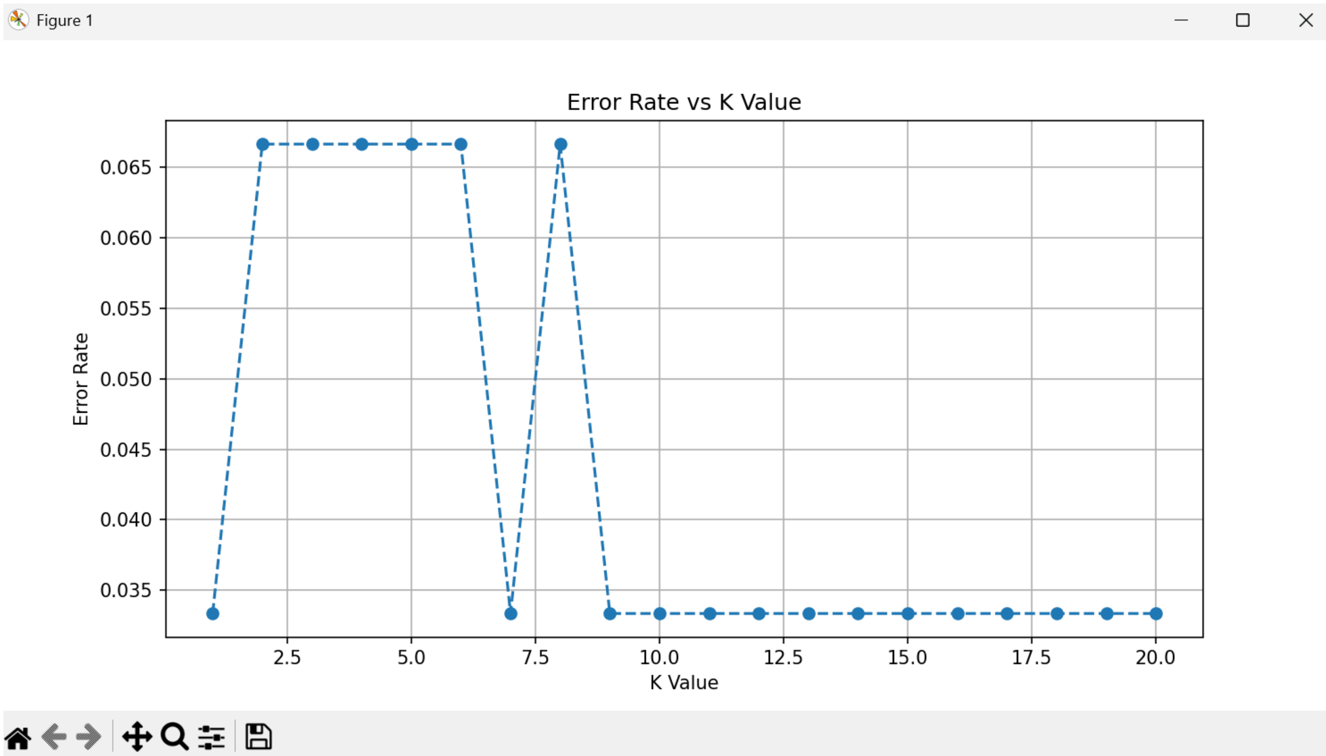
Figure 1
Error Rate vs K Value
Error Rate
K Value
0.065
0.060
0.055
0.050
0.045
0.040
0.035
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0